

Architecture, Vision and Delivery Status

A repository-backed explanation of what has been built, what remains, and the intended end game.

Executive alignment summary

PFA is no longer merely an FVG scanner. It is now a tested research and safety platform whose durable subject is the market itself: point-in-time market facts are captured once, then patterns, sequences, strategies, validation, sandbox behavior, and future agents are built as versioned annotations over that history.

How it started

The project did not begin as a designed visual product. It began with working analytical code and a simple request: turn this code into a visual app. The first interface appeared rapidly, then continued evolving in step with each new research and safety capability. That visual feedback loop is now part of the product method: the app makes the engine's evidence, limitations, and progress inspectable as the architecture grows.

Current proof	Meaning
29 implementation commits	A staged migration from regression protection through certification and training readiness.
24,595 tracked lines	22,665 C# lines, plus UI, JSON fixtures, and 1,567 lines of design/operations documentation.
221 tests passing	No failed or skipped tests; last verified build had zero warnings and zero errors.
29 commits awaiting push	The work is committed locally on pfa-market-intelligence; remote publication is still pending explicit destination approval.

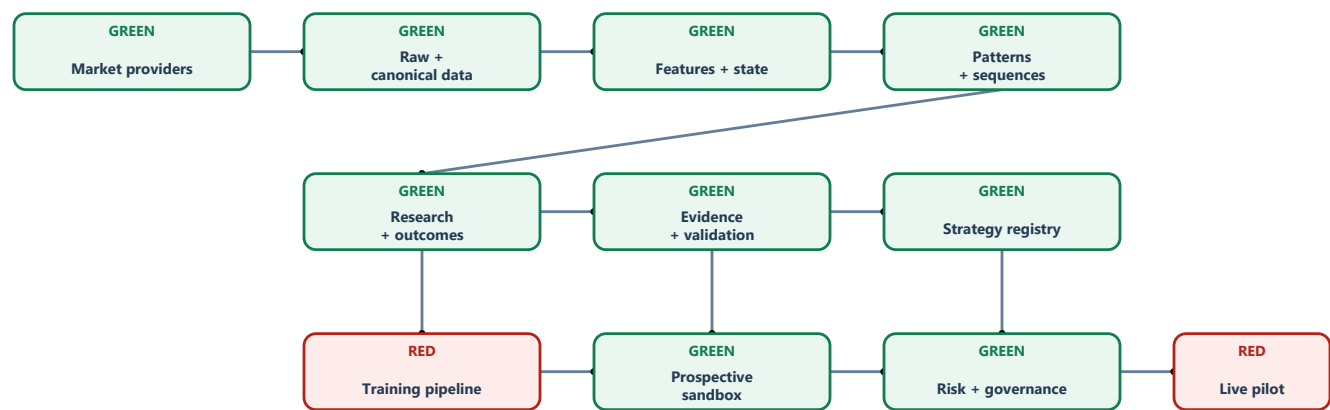
The key distinction

The architecture is broadly implemented; the evidence corpus and trained agent are not. We have built the factory, safety rails, and inspection stations. The next job is to run enough correctly versioned market data through that factory to produce trustworthy training examples.

End game

- Continuously capture reliable multi-market futures data with explicit contracts, sessions, provenance, revisions, and quality.
- Turn that timeline into reproducible state, feature, pattern, sequence, and outcome records without hindsight leakage.
- Train research agents on immutable temporal datasets and evaluate them on unseen, walk-forward, then prospective paper evidence.
- Help traders navigate prop-firm rules with calibrated decisions, including NO TRADE, while keeping billing separate from safety authority.
- Only after sustained evidence and explicit approval, permit a bounded MES paper/live pilot behind independent risk, reconciliation, credentials, audit, and kill-switch controls.

Architecture map: target and present state



Green means the subsystem exists in code and is covered by the current test suite.
Red means the end-state capability is not operationally complete, even if contracts or readiness gates exist.

The horizontal flow is intentional: data precedes interpretation; interpretation precedes evidence; evidence precedes authorization. The training pipeline consumes historical evidence but cannot promote itself. The live-pilot node remains red because readiness projections and certification simulations are not a functioning broker route.

Delivery roadmap



What has actually been built

Status	Capability	Concrete implementation
GREEN - IMPLEMENTED	Regression and compatibility	Golden masters preserve legacy FVG detection, aggregation, replay, scenarios, research, persistence, and version behavior.
GREEN - IMPLEMENTED	Market identity	Versioned futures instruments, contract identity, sessions, tick sizes, point values, and supported resolutions replace silent MES assumptions.
GREEN - IMPLEMENTED	Canonical timeline	Raw events and canonical bars carry timestamps, provenance, quality, revisions, corrections, and deterministic identity across live and backfill paths.
GREEN - IMPLEMENTED	Market intelligence	Point-in-time market state and feature definitions are separated from trading judgment and preserve KnownAtUtc semantics.
GREEN - IMPLEMENTED	Pattern modules	FVG, liquidity sweep, range breakout, and failed breakout detectors share generalized contracts and module registration.
GREEN - IMPLEMENTED	Universal evidence records	Observations, lifecycle, outcomes, metrics, events, lineage, and quality are stored independently of any one scanner.
GREEN - IMPLEMENTED	Sequence intelligence	Versioned definitions replay ordered, overlapping, partial, failed, and completed market behavior without rewriting source observations.
GREEN - IMPLEMENTED	Research and validation	General research, cross-day and cross-market evidence, explicit execution ambiguity, out-of-sample and walk-forward contracts are present.
GREEN - IMPLEMENTED	Historical operations	Durable, resumable historical jobs, windows, checkpoints, coverage, and a multi-asset campaign are represented.
GREEN - IMPLEMENTED	Order-flow foundation	Canonical order-flow events, classification, snapshots, persistence, and service boundaries exist; production source coverage remains separate.
GREEN - IMPLEMENTED	Sandbox and governance	Append-only virtual ledgers, conservative fills, risk policy, approval/veto decisions, emergency stop, incidents, and default-deny controls exist.
GREEN - IMPLEMENTED	Forward and machine research	Prospective campaign contracts, degradation monitoring, reproducible discovery runs, and non-privileged hypotheses exist.
GREEN - IMPLEMENTED	Certification	Immutable \$50K prop-firm rule packs and campaigns model latency, costs, drawdown, consistency, payout gates, and explicitly deny live routing.
GREEN - IMPLEMENTED	Product modularity	Core, Agent Research Lab, Live Agent, Advanced Strategies, coaching, and BYO-agent module contracts separate entitlement from safety.
GREEN - IMPLEMENTED	Agent readiness	Point-in-time training-example contracts, leakage rejection, dataset splits, and a readiness gate of 100 R labels over 90 days exist.

Code architecture by layer

API and product surfaces	Controllers expose data health, coverage, patterns, sequences, research, validation, sandbox, governance, certification, modules, and agent readiness. Browser screens support the market-intelligence view, sandbox decisions, certification, and Agent & Module Center.
Services and engines	Services orchestrate ingestion, aggregation, replay, detection, generic outcomes, sequence replay, research campaigns, scenario normalization, validation, sandbox operation, certification, governance, and training readiness.
Domain contracts	Versioned records define instruments, bars, features, patterns, outcomes, sequences, strategies, evidence, sandbox events, governance decisions, certification campaigns, product modules, and training examples.
Persistence	SQLite repositories preserve raw data, candles, canonical timeline, observations, outcomes, sequences, research runs, evidence, strategies, historical jobs, order flow, sandbox ledgers, governance, validation, discovery, and certification.
Tests	Golden-master and phase-specific tests assert deterministic identities, point-in-time chronology, idempotency, data isolation, ambiguity handling, nonactivation, safety gates, and database invariants.
Documentation	Architecture, migration, each phase, the research universe, agent lab, product UI, modular product model, owner direction, multi-asset campaign, and partner handoff are recorded as reviewable decisions.

Why this matters

The code is not 24,595 lines of trading logic. Most of its value is separation and traceability: the same market fact can be replayed through multiple detectors; a detector cannot silently become a strategy; a promising backtest cannot silently become an order; and a future agent cannot train on information that was unavailable at decision time.

Remaining work - red until proven complete

Status	Required capability	Definition of remaining work
RED - REQUIRED	Finish and verify the corpus	Complete the active multi-asset backfill; reconcile failures, gaps, duplicates, sparse contracts, provider limits, and authoritative database coverage.
RED - REQUIRED	Contract continuity policy	Approve versioned rollover selection, adjusted/unadjusted series behavior, and liquidity rules. Current dated contracts must not masquerade as continuous futures.
RED - REQUIRED	Session/calendar policy	Finalize CME trading dates, maintenance breaks, holidays, and early closes, then rebuild coverage and evidence under the approved version.
RED - REQUIRED	Outcome labeling at scale	Replay every active detector across supported timeframes, generate generic outcomes, then define strategy-specific entry/stop/target/R labels with costs and chronology.
RED - REQUIRED	Feature completeness	Add and validate market structure, displacement, session references, volume/volatility, and production-grade cross-market context where definition-pending.
RED - REQUIRED	Training dataset manifests	Materialize immutable train/validation/test manifests, frozen revisions, embargo rules, deduplication, leakage audit, and reproducible feature schemas.
RED - REQUIRED	Train baseline agents	Start with transparent baselines; measure calibration, expectancy after costs, drawdown, tail loss, stability, abstention quality, and regime degradation.
RED - REQUIRED	Model registry and reproducibility	Persist model code/data versions, hyperparameters, seed, artifacts, metrics, approvals, and immutable lineage. No model may self-promote.
RED - REQUIRED	Prospective paper operation	Run frozen strategies/models continuously in virtual accounts long enough to compare historical expectations with real forward behavior.
RED - REQUIRED	Production data operations	Add scheduler/service hosting, restart recovery, monitoring, alerts, backups, retention, archival, reconciliation, and capacity planning beyond local SQLite assumptions.
RED - REQUIRED	Identity, billing, and entitlement	Implement authenticated users, payment webhooks, server-side entitlements, signed partner assertions, revocation, audit, and least-privilege data scopes.
RED - REQUIRED	External module integration	Build and validate the Advanced Strategies API manifest, DTOs, compatibility, health, idempotency, security, and failure isolation.
RED - REQUIRED	Execution semantics	Approve fees, spread, slippage, latency, partial fills, order types, OCO ownership, cancel/replace, overlapping positions, and portfolio risk.
RED - REQUIRED	Broker paper certification	Select an officially supported provider; implement credential custody, broker-neutral routing, idempotency, reconciliation, rejected/partial fills, and failure drills.
RED - REQUIRED	Bounded live pilot approval	Only after stable walk-forward and nonzero forward evidence: explicit owner/governance authorization for MES, maximum one micro contract, with kill switch and incident ownership.

Immediate path to agent training

1	Close the data loop	Resume and finish the active 17-market dated-contract campaign. Treat database row counts and time coverage as authority, not provider-response counts.
2	Freeze semantic decisions	Version contract rollover, CME sessions, corrections, source priority, and feature definitions before creating a canonical training release.
3	Replay observations	Run FVG, liquidity sweep, range breakout, failed breakout, and approved sequences over every eligible instrument/timeframe partition.
4	Generate labels	Produce fixed-horizon generic outcomes first. Add strategy-specific R labels only from frozen execution hypotheses with explicit costs and ambiguity handling.
5	Publish Dataset V1	Create an immutable manifest: IDs, feature schema, time bounds, revisions, exclusions, hashes, label availability, and temporal splits.
6	Train transparent baselines	Benchmark simple rules, logistic/linear models, and calibrated tree models before complex agents. A model must beat honest baselines after costs.
7	Walk forward	Evaluate untouched folds and regimes. Track calibration, net expectancy, maximum drawdown, tail loss, turnover, cost sensitivity, and NO TRADE behavior.
8	Prospective sandbox	Freeze the winning candidate and operate it without retraining on its evaluation period. Compare expected and realized distributions.

Earliest honest training start

Training can begin as soon as a first immutable, temporally valid dataset contains enough labels to support a meaningful split. The current code gate uses at least 100 distinct R-labeled outcomes spanning 90 days. That is a research eligibility floor, not a claim of adequacy, profitability, or permission to trade.

Phase-by-phase delivery ledger

Phase	Deliverable	State
0	Golden masters	GREEN - IMPLEMENTED
1	Instruments, sessions, contracts	GREEN - IMPLEMENTED
2	Canonical timeline	GREEN - IMPLEMENTED
3	Market state and features	GREEN - IMPLEMENTED
4	Pattern contracts	GREEN - IMPLEMENTED
5	FVG module adapter	GREEN - IMPLEMENTED
6	Universal observations/outcomes	GREEN - IMPLEMENTED
7	Sequence intelligence	GREEN - IMPLEMENTED
8	Liquidity sweep	GREEN - IMPLEMENTED
9	Breakout modules	GREEN - IMPLEMENTED
10	Strategy registry	GREEN - IMPLEMENTED
11	General research	GREEN - IMPLEMENTED
12	Cross-day evidence	GREEN - IMPLEMENTED
13	Cross-market evidence	GREEN - IMPLEMENTED
14	Ambiguity escalation	GREEN - IMPLEMENTED
15	Historical pipeline	GREEN - IMPLEMENTED
16	Walk-forward validation	GREEN - IMPLEMENTED
17	Order-flow foundation	GREEN - IMPLEMENTED
18	Virtual sandbox	GREEN - IMPLEMENTED
19	Risk and governance	GREEN - IMPLEMENTED
20	Forward campaigns	GREEN - IMPLEMENTED
21	Machine discovery	GREEN - IMPLEMENTED
22	Certification/readiness infrastructure	GREEN - IMPLEMENTED
23	Operational corpus completion	RED - REQUIRED
24	Agent dataset + baseline training	RED - REQUIRED
25	Sustained forward paper evidence	RED - REQUIRED
26	Broker-certified bounded pilot	RED - REQUIRED

Vision decisions for owner review

Market scope

Is the 17-root futures universe the right initial research boundary, or should the first training release narrow to MES/MNQ plus a small context set?

Agent job

Should the first agent predict market outcomes, rank setups, recommend NO TRADE, coach prop-firm behavior, or combine these as separately evaluated heads?

Primary objective

Which objective governs model selection: net expectancy after costs, drawdown control, challenge-pass probability, payout probability, calibration, or a constrained combination?

Time horizon

What decision cadence and holding horizons define the first trainable product: intraday 1m/5m, higher-timeframe context, or both?

Rollover

Should training use explicit dated contracts only, a versioned unadjusted continuous series, adjusted continuous history, or multiple views?

Evidence threshold

What minimum sample, date span, regimes, walk-forward folds, and forward-paper duration are required before a candidate advances?

Product boundary

Is the core commercial value intelligence and coaching, a research agent, a live assistant, partner strategies, or a modular bundle with distinct safety gates?

Pilot authority

Confirm that MES, one micro contract maximum, and paper-first remain hard ceilings until a separately documented live approval.

Alignment test	If the intended vision differs on the agent's job, target customer, primary objective, or live-trading boundary, revise those decisions before scaling data collection or training. Those choices determine labels, datasets, evaluation, UI, and governance.
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Safety invariants that should not change

- Patterns describe market behavior; they are not strategies.
- Training labels and outcomes cannot appear in predictors before they were knowable.
- A paid entitlement grants product access, never risk approval or broker authority.
- Research, validation, walk-forward, and prospective paper periods remain separate populations.
- Ambiguous execution is unresolved or conservative; it is never optimistically backfilled.
- NO TRADE is a first-class decision and must be evaluated.
- Discovery proposes; evidence tests; governance promotes. The agent cannot promote itself.
- Every material dataset, feature, pattern, strategy, fill model, rule pack, and model is versioned and reproducible.
- Dated futures contracts remain explicit unless a reviewed rollover/adjustment policy creates a continuous series.
- Live execution remains absent until official provider support, credentials, reconciliation, duplicate protection, failure drills, independent governance, and explicit owner approval all exist.

Repository evidence used

This report was derived from the current branch history through commit aacb7ba, the complete tracked source tree, passing build/test results, Architecture V1, Migration Plan V1, Agent Research Lab V1, phase documents, modular product architecture, multi-asset campaign plan, owner direction, and the Advanced Strategies handoff. Status colors indicate implementation presence and verification, not production readiness or profitability.

Recommended owner decision

Approve or revise the end-game statement and the eight vision decisions in this report. Once aligned, the immediate engineering priority is corpus completion -> immutable Dataset V1 -> baseline training -> walk-forward evidence -> prospective paper operation.